When Zero-Day Becomes Training Data: Learning-Curve Stability under Out-of-Distribution and Domain Adaptation on the MalNet-Tiny datasets

Itamar Link
itamarlink@mail.tau.ac.il

Shai Habi
shaihabi@mail.tau.ac.il

Tomer Gal Netser
galnetser@mail.tau.ac.il

Abstract

Malware detection systems trained on known threats ("one-days") must also recognize new, unseen malware families ("zero-days"). We study out-of-distribution (OOD) detection and domain adaptation in graph neural networks (GNNs), asking how many labeled examples from a zero-day family are needed to detect future instances. Using function call graphs (FCGs) from MalNet-Tiny, Common, and Distinct, we compare GCN, GIN, GAT, and a graph transformer (GT) as labeled zero-day samples accumulate, measuring detection performance and learning-curve stability. Across four analyzed families, training with benign graphs and 25 labeled examples per zero-day family yields mean held-out zero-day recall of 84–97% across models. When learning alongside known malware, GT achieves the highest mean normalized area under the zero-day recall learning curve (0.80 versus 0.47–0.54 for the other models).

1 Introduction

We refer to known malware families as **one-days** and families absent from training as **zero-days**.

One approach to malware detection with graph neural networks (GNNs) uses **Function Call Graphs (FCGs)**, where nodes represent program functions and edges represent calls between them [2, 3].

Freitas et al. [2] introduced MalNet, containing more than 1.2 million FCGs extracted from Android applications, and its 5,000-sample MalNet-Tiny subset. These resources supported subsequent graph-based malware research, including the Common and Distinct datasets introduced by Tran et al. [3], on which we build our experiments.

1.1 *Motivation and Research Question*

GNNs perform well on MalNet-Tiny when training and test data share a distribution, but performance declines under out-of-distribution (OOD) conditions [3]. To our knowledge, low-shot domain adaptation, where only a few labeled samples from a new malware family are available, remains insufficiently studied in GNN-based malware detection.

This two settings are particularly relevant to cybersecurity, where detection systems are trained on previously discovered malware ("one-days") but are also expected to detect new unseen malware ("zero-day"). The training data for these systems always involved, and is provided with new emerged malware family that were correctly detected. These two vectors motivate our research question, which is **how many samples from a newly encountered malware family, do GNNs require in order to reliably detect future instances from that family?**

Our complete pipeline and results are available in our GitHub repository.

1.2 *Datasets*

We use three datasets of 5,000 samples each. Throughout this paper, we refer to MalNet-Tiny [2] as **Original**. It contains four malware families and a benign class, with 1,000 samples per class. **Common** [3] contains 1,000 new samples from each of the same classes; **Distinct** [3] contains five unseen malware families, with 1,000 samples each and no benign graphs.

Together, they contain 2,000 benign and 13,000 malicious samples across nine malware types; these are sample counts, not counts of unique structures. Each class has a 70/10/20 training/validation/test split. Table 1 reports mean graph sizes.

2 Literature Review

Freitas et al. [2] introduced MalNet to benchmark graph representation methods on large, diverse and imbalanced collections of software graphs. Their MalNet-Tiny experiment evaluated supervised five-class classification: assigning each FCG to the benign class or one of four malware types, with all

Family	Avg. Nodes	Avg. Edges
Original		
Addisplay	1631.22	2920.03
Adware	2506.74	5597.27
Benign	2064.03	3886.76
Downloader	49.38	55.31
Trojan	799.87	1840.32
Common		
Addisplay	2438.15	4836.79
Adware	4159.36	7990.85
Benign	1997.20	3729.22
Downloader	683.17	1882.70
Trojan	1863.02	4247.45
Distinct		
Clicker++Trojan	3806.42	9109.92
Malware	1185.28	2491.26
Riskware	2852.60	6688.56
SPR	4439.16	6376.55
Spyware	1427.86	2784.22

Table 1: Average graph structural size.

five classes represented during training. Using stratified 70/10/20 training/validation/test splits, GIN and GCN achieved test accuracies of 90% and 81%, respectively. These percentages measure correct class assignments within the benchmark's distribution; they do not measure detection of a malware family absent from training.

The authors also identified imbalanced graph classification as an underexplored research direction. Although MalNet-Tiny is balanced across its five classes, merging the malware types into one binary class produces four malicious graphs per benign graph. Our Per-Family setting explores the complementary situation in which benign examples outnumber the available zero-day examples, and we report macro-F1 alongside class-sensitive metrics. To broaden evaluation beyond MalNet-Tiny's limited 5,000 samples, we incorporate Common and Distinct, tripling the total sample count to 15,000 and doubling the number of benign samples from 1,000 to 2,000.

Tran et al. [3] investigated whether semantic enrichment of FCGs improves robustness to distribution shift. Unlike the released Original graphs, which contain no node attributes, their enriched representations incorporate function metadata and code embeddings. Their Common experiment trains on Original and evaluates different malware variants under the same five class labels. With Local Degree Profile (LDP), a structural representation summarizing node and neighborhood degrees, mean five-class accuracy drops from approximately 91% on Original to 48% on Common across GCN, GIN, GPS and Exphormer. Thus, high accuracy on Original does not establish reliable classification of new variants. They found that transformer-based architectures achieved higher in-distribution accuracy than message-passing GNNs, but this advantage did not persist under distribution shift. With semantic features, message-passing GNNs nearly matched the transformers' in-distribution accuracy and outperformed them on Common.

Their Distinct experiments study adaptation to five new malware classes using labeled target examples, comparing full fine-tuning, AdapterGNN and nearest-neighbor classification of graph embeddings. For fine-tuning-based methods, the classification head is reinitialized for the new labels. This evaluates learning to distinguish the new classes after adaptation, rather than direct transfer of the original classifier.

Our shared benign-versus-malware label space makes direct evaluation on unseen families possible: a zero-day graph remains malicious even when its family was absent from training. We then vary the number of labeled zero-day examples to measure how detection improves and when learning curves stabilize. For these experiments, we use node in-degree and out-degree as a common structural representation computable from all three datasets. This keeps feature construction consistent as labeled zero-day samples accumulate, without requiring reconstruction of semantic attributes.

Bilot et al. [1] review graph representations and learning methods for malware detection, including challenges from adversarial attacks. Wu et al. [4] distinguish graph domain adaptation, OOD generalization and continual learning according to the data and supervision available. These perspectives motivate separating detection before any zero-day examples are available from learning as labeled examples accumulate.

Finally, to the best of our knowledge, our method for evaluating Original, Common and Distinct as binary classification under OOD and domain adaptation conditions is novel.

3 Methodology

We compare GCN, GIN, GAT, and GT for benign-versus-malware classification.

3.1 Shared Experimental Configuration

We use predefined values $k \in \{0, 25, 50, 100, 175, 250, 350, 500, 700\}$, where k

is the number of labeled samples from the zero-day family added to training. Denser sampling at low k captures potentially larger gains from early examples. We shuffle each family split once during preprocessing and use the first k training samples of the zero-day family, so larger subsets retain all earlier examples. Given the limited data, we combine each zero-day family's validation and test splits, leaving 700 training samples and 300 held-out samples. **We used the same configurations that outperform in the entire MalNet dataset** [2]. we used five GNN layers and Adam optimizer as the MalNet authors, and we adopted the hidden dimension of 64 that performed best in their experiments on the entire MalNet dataset. Their best learning rate was 0.0001, but we used 0.001, another value they evaluated, for our smaller training sets. Tran et al. [3] did not report their configurations for transformer-based GNNs, therefore we chose 4 as the number of attention heads for both GAT and GT, choosing a moderate model size for our small training sets.

Motivated by the findings of Tran et al. [3] discussed in Section 2, we use node in-degree and out-degree features, applying log1p to reduce the influence of very large degrees.

To account for class imbalance, BCEWithLogitsLoss uses a positive-class weight equal to the benign-to-malware sample ratio in each training set.

We measure macro-F1 on the full test set and zero-day accuracy (recall) on the held-out samples of the zero-day family. We emphasize that **all experimental parameters were fixed in advance and were not tuned on our data.**

3.2 Experimental Settings

Our experiments simulate zero-day discovery: a family may initially escape detection, while increasing k represents labeled examples accumulating as it is discovered and investigated. OOD models detection alongside known malware; Per-Family isolates learning from benign and zero-day graphs alone. Both settings examine when additional examples yield stable detection on held-out samples.

Dataset Analysis: Models trained on Original are evaluated on Common and on Distinct plus Common's benign graphs, establishing transfer performance before exposure to any Distinct family.

Out-of-Distribution Learning Analysis: We treat each Distinct family as a zero-day family. For each family and k, we train a fresh model of each architecture on Original's 5,000 graphs plus k zero-day samples. Testing uses Common's 5,000 graphs plus the zero-day family's 300 held-out samples.

Per-Family Analysis: Training uses 1,400 benign graphs plus k samples from the zero-day family; testing uses 600 benign graphs and the zero-day family's 300 held-out samples, isolating learning from the other malware types. Both splits contain more benign than malicious graphs, reflecting benign software's predominance in deployment.

4 Results and Discussion

We exclude SPR from family-level comparisons based on the duplication concern illustrated by its Per-Family curves in Figure 1; Section 4.2 explains this decision in detail. Each run's binary and macro-F1, with its test-set class counts, determine a unique integer confusion matrix, so we also report the benign false-positive rate (FPR) and balanced accuracy (BA). The Common Dataset Analysis run and the four retained OOD runs at $k = 0$ repeat one training configuration and seed, yet vary (presumably through non-deterministic GPU operations), with a sample standard deviation of macro-F1 on Common of up to 0.03.

4.1 Dataset Analysis

	ID	Shift ($k = 0$)			
Model	**Per-Family**	**Common**	**Distinct**	**FPR**	**ALC**
GCN	.79 / .83	**.52** / .61	**.41** / .53	.32	.54
GIN	.87 / .87	.44 / .64	.28 / .55	.04	.47
GAT	.80 / .82	.49 / .66	.29 / .56	.05	.51
GT	**.89** / **.90**	.51 / **.68**	.35 / **.60**	.05	**.80**

Table 2: Macro-F1 / BA in distribution (Per-Family, mean over the four retained families and $k > 0$) and under shift (Dataset Analysis; full Distinct, including SPR, plus Common's benign graphs). FPR: benign false-positive rate on Common. ALC: trapezoidal area under OOD zero-day recall versus linear k, divided by 700 and averaged across four retained families.

Table 2 shows poor transfer from Original in macro-F1 and BA. Malware recall is 0.31–0.53 on Common and 0.14–0.39 on the full Distinct dataset. GIN, GAT and GT keep the benign FPR at about 4–5%, whereas GCN also flags about 30% of benign graphs across the retained $k = 0$ runs. We hypothesize a size shortcut: Common's malware types are 1.5–14 times larger than in Original (Table 1), while benign graphs are not, and the mean zero-shot accuracy of GCN, GIN, and GAT on the four

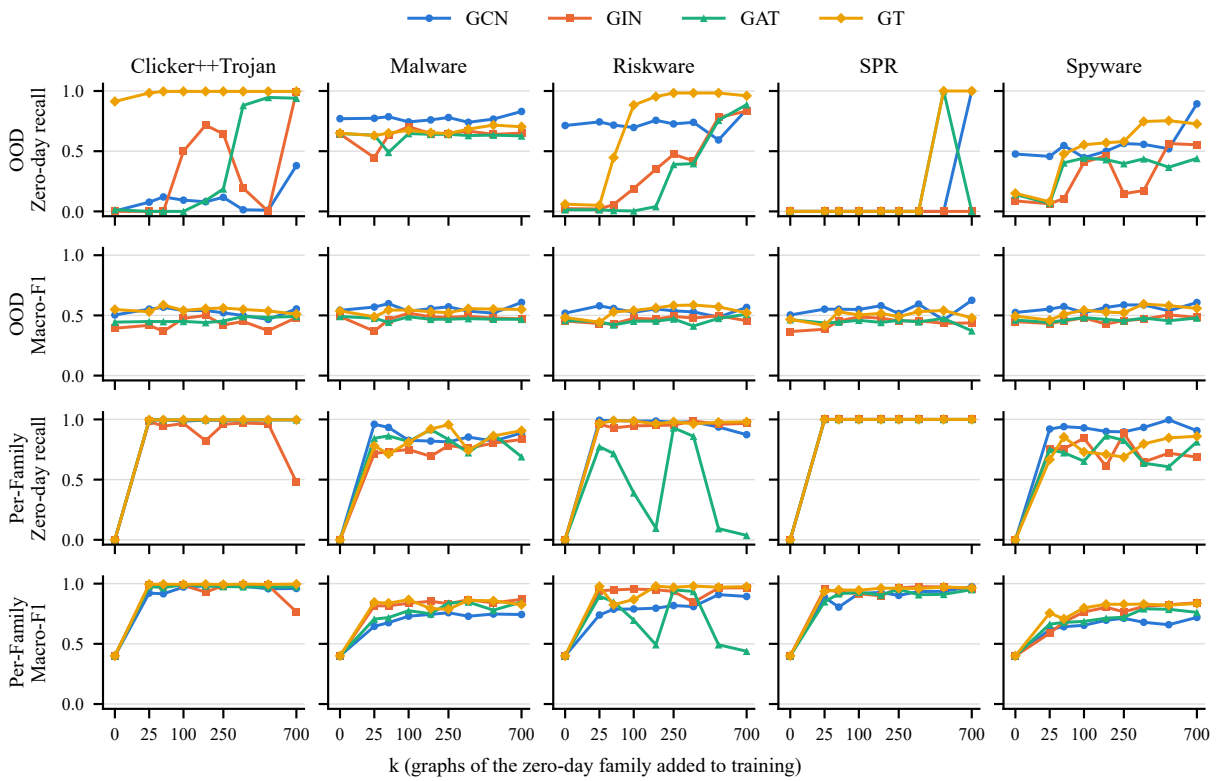


Figure 1: Zero-day recall (300 held-out family samples) and macro-F1 (full test set) vs. k zero-day training samples (square-root scale). Top two rows: OOD; bottom two rows: Per-Family. Each pair shows recall above macro-F1.

retained Distinct families is negatively associated with mean node count (Spearman $\rho = -0.8$), from 69% on *Malware*, the smallest, to less than 1% on *Clicker++Trojan*, the largest. With only four families, this association is descriptive evidence for the hypothesis.

4.2 Learning Curves

SPR as a diagnostic case. SPR's Per-Family recall rises from 0 to 1 at $k = 25$ for all models and remains there (Figure 1). Its 1,000 samples contain only nine distinct structures, with duplicates shared across training and held-out splits. This saturation is consistent with recognising repeated graphs rather than generalisation. We show SPR diagnostically but exclude it from family-level comparisons.

Family	GCN	GIN	GAT	GT
Clicker++Trojan	–/100	–/–	0/25	0/25
Malware	–/100	100/100	0/–	0/350
Riskware	100/500	0/500	500/–	100/175
Spyware	175/175	250/175	0/350	350/100

Table 3: Convergence point k^* of each retained family's macro-F1 learning curve, OOD / Per-Family: the smallest k after which every later run's macro-F1 lies within $\delta = 0.05$ of its value at k^*. –: not converged before $k = 700$.

We retain $\delta = 0.05$, above the largest standard deviation across retained $k = 0$ repeats (0.03), to allow run-to-run variation. Convergence requires a later run, so $k = 700$ cannot qualify. **In the Per-Family setting**, 25 samples raise zero-day accuracy averaged over the four retained families from 0 to 0.84–0.97 across models (Figure 1), and macro-F1 converges for 13 of the 16 family–model pairs. GT and GCN converge on all four families and tie for the lowest median $k^* = 137.5$, although GT converges by $k = 350$ on every family, compared with $k = 500$ for GCN. GAT does not converge on *Malware* and *Riskware*, and GIN not on *Clicker++Trojan*, whose macro-F1 falls from 0.99 at $k = 500$ to 0.77 at $k = 700$.

In the OOD setting, where the family is only 5.7% of the test set, macro-F1 changes little with k: 13 of the 16 curves converge, nine of them by $k = 100$. GCN converges on only two of the four families. This overall stability hides the family-level learning in Figure 1: adding 25 samples changes family-averaged zero-day accuracy by between -0.06 and $+0.02$ across models. Over the full grid, GT detects the new family best (ALC 0.80 vs. 0.47–0.54 for the MPNNs, Table 2; e.g., *Riskware*: 0.06 at $k = 0$, 0.98 at $k = 250$), while

GIN shows sharp drops (mean maximal drawdown 0.32, vs. 0.03 for GT).

4.3 Discussion

Setting and family. At $k = 25$, the positive loss weight is $1{,}400/25 = 56$ in Per-Family versus about 0.25 in OOD, where the family comprises at most 15% of malware. Among other setting differences, this 225-fold contrast may contribute to slower OOD gains in zero-day accuracy.

Comparison with Tran et al. [3]. They found that transformer-based architectures achieve higher in-distribution accuracy than MPNNs but perform worse on distribution-shifted data. In our binary setting, the first half holds but the second does not. GT has the highest mean Per-Family (in-distribution) macro-F1 and BA, 0.89 and 0.90, respectively, and the lowest training loss in 66 of 72 matched family-learning conditions. Under shift, it keeps the highest BA on both Dataset Analysis test sets (Common: 0.68, above every other model in all five retained repeat runs; full Distinct: 0.60) and on average over the retained OOD conditions; BA, like accuracy on their class-balanced test sets, weights every class equally. Only macro-F1 ranks GCN first under shift: its higher malware recall comes with about 30% benign false positives, a trade-off that macro-F1 rewards on test sets that are about 80% malware. Our binary task, degree features and neighbour-only attention (unlike GPS and Exphormer) may explain the discrepancy, but this is untested.

GAT failure. GAT has the highest Per-Family mean maximal drawdown (0.34) and largest individual drop (*Riskware*: 0.93 at $k = 250$ to 0.04 at $k = 700$); GT has the same four heads without comparable collapse. GAT uses an attention-weighted neighbourhood average, including self-loops; GT adds a separate root term. A Per-Family *Riskware* ablation ($k = 250$, 500 and 700; three seeds each) partly supports this explanation: removing GT's root term reduced zero-day recall in all nine pairs (mean 0.97 to 0.72). Adding one to GAT increased mean recall (0.49 to 0.79) and reduced collapses (recall below 0.5) from five of nine runs to two, but improved only five pairs. Collapsed runs suggest representation collapse more than within-graph over-smoothing: mean cosine similarity is 0.76 versus 0.24 in healthy runs for pooled graph embeddings, and 0.45 versus 0.28 for last-layer node embeddings; most *Riskware* graphs are labelled benign. GAT collapsed for some seeds but not others at every tested k, so its single-seed drops (Figure 1) cannot be attributed to k alone.

Limitations. Each learning-curve point is a single 30-epoch run without validation, evaluated at a 0.5 threshold. Structural-only features and the full dataset's 2,000 benign versus 13,000 malicious samples further limit applicability to deployments, where benign software dominates.

5 Conclusion

We asked how many labeled samples from a zero-day malware family GNNs need for reliable detection. Across the four retained families, 25 samples raise mean Per-Family zero-day accuracy to 0.84–0.97 across models. Across the tested training sizes, macro-F1 converges within $\delta = 0.05$ for 13 of 16 family–model pairs, with GT and GCN converging on all four families. When learning alongside known malware, overall macro-F1 changes little with sample count, but GT detects the zero-day family best across the grid and maintains the highest balanced accuracy under distribution shift.

Structural duplication limits interpretation of these findings as generalisation. We are considering asking the original authors to release additional data to support evaluation with structurally disjoint training and test sets. Future work should also test family-specific weighting and include more representative benign data.

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